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Thalassos — Pitch Deck Research & Storytelling Framework

What this is: A research-backed guide for building your investor pitch deck. Not slides — a **framework** for what to include, how to structure the story, and what VCs/judges actually care about. Use this as your blueprint when you sit down to design.

Part 1: What the Research Says

What Elite Early-Stage Decks Have in Common

Based on Sequoia's framework, Y Combinator principles, and deep tech / climate tech pitch best practices:

1. **A narrative spine, not a data dump.** The best decks tell a story that builds tension, then resolves it. Judges spend <3 minutes on a first look — if slide 2 doesn't hook them, they're gone.
2. **Problem as human pain first, then scale.** Don't open with market size. Open with a *person* suffering. Then zoom out to show it's systemic.
3. **"Why Now?" framing.** Deep tech investors need to know why this moment in history makes your solution possible. What changed?
4. **De-risking milestones over traction.** Pre-seed companies don't have revenue. Show what you've *proven* — lab results, validated models, expert confirmations, cross-geography testing.
5. **Bottom-up TAM.** Don't say "\$15B market." Build it from: number of wells × cost per screening × countries. Judges trust math, not Statista links.
6. **Team = founder-market fit.** Why are *you two* the ones to solve *this* problem? Not just credentials — the obsession.
7. **Vision that's 10x bigger than the product.** The product is arsenic screening. The vision is *planetary hydrogeological intelligence*. End with the vision.

What Moonshot Judges Specifically Look For

This isn't a normal startup pitch. Moonshot = audacious, world-scale, long-horizon. Judges want:

- **Scale of ambition** — Is this a billion-person problem?
- **Technical defensibility** — Can incumbents copy this in 6 months?
- **Path from science → deployment** — Is this a paper or a product?
- **Founder conviction** — Do these people *burn* for this problem?

TKS Playbook Principles (How to Present)

Principle	What It Means for Your Deck
Emotion	Slide 2 should make judges <i>feel</i> something. Sadness, anger, urgency. Not just "200M people affected" — tell the story of <i>one mother</i> .
Storytelling	The whole deck is one story: world is broken → here's why → we fixed it → here's proof → here's where we're going. Never break the arc.
Boss Mentality	Don't hedge. "180x cheaper" not "approximately more affordable." "98.07% accuracy" not "we think it's pretty accurate." Own your numbers.
Authenticity	You're 15 and 17. Don't pretend to be 30. Lean into it. Two teenagers building planetary infrastructure is a <i>better</i> story than two MBA grads.
Legit Content	56,952 validated well sites. Cross-geography proof. BGS hydrogeologist confirmation. This is not hand-waving. Show the receipts.
Enthusiasm	When you hit the Vision slide, your voice should change. This is where you show judges the world you see. Make them <i>want</i> to live in it.

Part 2: The Narrative Arc (14 Slides)

The Story Structure

ACT 1: THE WORLD IS BROKEN (Slides 1-4)

Title → Human Story → Scale of Crisis → Why Nothing Works

ACT 2: WE FIXED IT (Slides 5-7)

Solution → Why We're Better → Technical Proof

ACT 3: THIS IS A BUSINESS (Slides 8-10)

Market → Revenue Model → Where We Are Now

ACT 4: JOIN US (Slides 11-14)

Team → The Ask → Vision → Close

Slide-by-Slide Framework

SLIDE 1 — Title

Purpose: Set the tone. Institutional, serious, not playful.

Content direction:

- Company name: THALASSOS
- Tagline: "Planetary Hydrogeological Intelligence"
- One-liner: "We predict which wells are toxic before anyone drinks from them."
- Clean, dark background. No clutter.

Design note: Think Nature journal cover, not Canva startup template.

SLIDE 2 — The Human Story (THE HOOK)

Purpose: Emotional gut-punch. This is the single most important slide in the deck.

Content direction: Use the story from your comparison doc verbatim:

"In rural Bangladesh, a mother draws water from the family tube well installed twenty years ago. The water looks clear. It tastes normal. Her children have drunk it daily since birth. No one told her that the well fifty metres away tested at 400 µg/L of arsenic — forty times the WHO limit. Her well was never tested because the budget ran out after sampling two hundred wells in a district of fifty thousand."

How to present this slide:

- Minimal text on screen. Maybe just the quote or a single image.
- YOU speak the story. Don't read from the slide.
- **Pause after.** Let silence hang for 2-3 seconds.
- Then say: *"This is not negligence. This is mathematics."*
- Transition to next slide.

[!TIP] This follows Eric Edmeades' technique and Peter Diamandis' "emotional intro" — hook the audience emotionally before you give them data.

SLIDE 3 — The Crisis at Scale (CONSEQUENCES)

Purpose: Zoom out from one mother to 200 million people. Make it feel systemic and urgent.

Key data points to include (from your comparison doc):

Metric	Value	Source
People affected globally	200 million	Podgorski & Berg, Science 2020
Countries with contamination	70+	WHO/UNICEF JMP
Tube wells in Bangladesh	10 million	BGS-DPHE National Survey
Cost to test them all	500M–2 billion	Calculated
Years since last full survey	36 years (1990)	BGS
Wells contaminated in that survey	41.9%	BGS-DPHE dataset
Economic burden	>\$10 billion/year	Int'l Water & Sanitation Centre

Spoken framing: "In 1990, the British Geological Survey tested Bangladesh's groundwater for the first and last time. They found 41.9% of wells poisoned. That was 36 years ago. The geology hasn't changed. But neither has the data."

SLIDE 4 — Why Nothing Has Worked (ROOT CAUSE)

Purpose: Prove you understand the problem at a deeper level than anyone else. This is "legit content."

The causal chain (present visually as a cascade):

1. **People are being poisoned** → They consume toxic groundwater without knowing — symptoms appear after years
2. **They remain unaware** → Universal testing is never conducted before consumption
3. **Testing is avoided** → 50–200/sample, centralized labs don't exist at rural scale
4. **Spot-testing fails** → Contamination is non-linear: toxic well can be 100m from a safe one
5. **ROOT CAUSE:** The global paradigm treats geology as secondary. Nobody reads the mathematical signatures the earth broadcasts.

Transition line (spoken): *"Everyone has been trying to test the water. We read the earth instead."*

[!IMPORTANT] This is your "Why Now?" moment. The insight is: geology contains predictive signatures that existing methods ignore. You're not just cheaper — you're a fundamentally different approach.

SLIDE 5 — The Solution (TWO-PHASE ARCHITECTURE)

Purpose: Show how it works. Keep it visual, not text-heavy.

Content framework:

Phase 1 — Satellite Intelligence (Zero Cost)

- 41 parameters from NASA satellites + public geology data
- AI produces country-wide risk map
- 10ms inference per well
- 86% accuracy at \$0 per well
- Eliminates need to physically visit 80% of sites

Phase 2 — Ground Verification (\$1.50)

- \$1.50 multiplexed paper strip (5 parameters)
- Community health worker dips, waits 30s, photographs
- Smartphone app reads result instantly

- Combined accuracy: 98.07%
- Only deployed to top 15-20% flagged wells

Feedback Loop: Every Phase 2 result feeds back into Phase 1 → accuracy climbs over time (86% → 90% → 93%+)

Design direction: Visual flow diagram. Satellite → AI Model → Risk Map → Flagged Wells → Dip Strip → Data Feeds Back. Show, don't tell.

SLIDE 6 — Why We're Better (COMPARISON)

Purpose: Hard numbers that prove superiority. Boss mentality — no hedging.

Comparison table (from your docs):

	Traditional Lab	Thalassos
Cost per well	50–200	0+1.50
Time	Weeks to months	Minutes
Coverage	Budget exhausted	Country-wide
Skilled labor	Lab technicians	Health workers
Accuracy	Varies	98.07%
Scalability	Linear cost	Near-zero marginal

vs. Published academic methods (this is your credibility bomb):

Best Prior Work (Podgorski 2020)	Thalassos
80,000 samples	56,952 samples
0.82 AUC	0.9807 AUC
\$100/test	0–1.50/test
Single method	Two-phase hybrid

Key line: "180× cheaper. Hours instead of months. More accurate than most wet labs."

SLIDE 7 — Technical Depth (THE SCIENCE)

Purpose: Prove this isn't hand-waving for technical judges. Show the mathematical engine.

Framework for content:

Core innovation: You don't predict arsenic from chemistry. You detect spatial "fingerprints" left by geological processes.

Mathematical toolkit:

- Topological Data Analysis (TDA) — RIPSER algorithm for "chemical fingerprints" in well clustering patterns
- Information Geometry — Fisher information, KL divergence for anomaly detection
- Spatial Graph Theory — PageRank, betweenness centrality on contamination flow networks
- XGBoost ensemble across 73 total parameters (41 remote + 5 strip + 27 derived)

Validation proof points:

- 56,952 real well sites (USGS + BGS)
- Cross-geography: US (5.7% contamination) AND Bangladesh (41.9%) — completely different geologies, both work
- Spatial 5-fold cross-validation: 0.982 ± 0.003 AUC
- Conformal prediction: 95.7% coverage
- Exceeds government threshold (91%) by 7.07 percentage points
- BGS hydrogeologist confirmation

[!TIP] Don't try to explain TDA on stage. Say: "We use topological data analysis — a branch of pure mathematics — to find 'chemical fingerprints' hidden in the spatial patterns of well contamination." That's it. Technical judges will know what TDA is. Non-technical judges get the metaphor.

SLIDE 8 — Market Opportunity (ECONOMICS)

Purpose: Show this is a massive market with clear revenue potential.

TAM/SAM/SOM (bottom-up from your comparison doc):

Segment	Market Size
Arsenic testing globally	\$2–4B

Segment	Market Size
Fluoride testing	\$1–2B
Mining water risk / due diligence	\$3B
Agricultural water quality	\$5B
Total TAM	\$11.5–15B
Thalassos-addressable (PHI platform)	\$2.2–4.4B/year

Beachhead strategy: Bangladesh (10M wells) → India (Ganges-Brahmaputra) → Vietnam (Mekong Delta) → Global platform

Why this is believable at pre-seed: The arsenic test strip market alone is ~\$1.8B in 2026 (CAGR 6.5%). Asia-Pacific = 38.6% share. You're entering the biggest segment of the biggest region.

SLIDE 9 — Business Model (HOW YOU MAKE MONEY)

Purpose: Show multiple revenue streams and a path to \$80M revenue.

Five streams (from comparison doc):

1. **Government Contracts (40%)** — National risk mapping: 0.10–0.30/well.
Bangladesh 10M wells = \$1–3M.
2. **NGO/Development Grants (25%)** — Gates Foundation (5–15M), *USAID* (2–5M), World Bank (10–50M), *UNICEF* (3–10M)
3. **Enterprise: Mining & Ag (20%)** — Site water risk: 10K–50K/site. Targets: Rio Tinto, Nestlé, Unilever.
4. **Insurance (10%)** — Parametric water quality insurance with Swiss Re, Munich Re.
5. **Data & API Licensing (5%)** — Risk map API, white-label platform for governments.

5-year trajectory (from comparison doc):

Year	Revenue	Wells Screened	Countries
1	\$2M	2M	2
3	\$20M	25M	10

Year	Revenue	Wells Screened	Countries
5	\$80M	100M	20

SLIDE 10 — Where We Are Now (DE-RISKING)

Purpose: Since you're pre-seed/ideation, frame everything as milestones that prove the science is real.

What you've achieved:

-  ML architecture designed, trained, validated on 56,952 wells
-  Cross-geography generalization proven (US + Bangladesh)
-  Validated by hydrogeologist at British Geological Survey
-  Academic interest from Imperial College London
-  Y Combinator Startup School India participants
-  Comprehensive technical paper written
-  10,000-test side-by-side validation in progress

What this funding unlocks:

- Complete validation study
 - Legal, admin, operational setup
 - Build awareness — put solution in front of governments/NGOs
 - Begin Bangladesh pilot conversations
-

SLIDE 11 — The Team (WHY US)

Purpose: Founder-market fit. Why you two are uniquely qualified.

Dhruvin Bhandari — Co-Founder & CEO

- Designed the ML architecture. Trained on 56,952 wells.
- Proved cross-continent generalization via physics-informed AI
- Lodha Genius Fellow @ Ashoka University
- 2nd / 5,000+ at Masters Union National AI Hackathon
- Research Intern @ CSIR India | TKS Innovator '26

Taha Usman — Co-Founder & CTO

- 15 years old. Shipped 5 SaaS products.
- Stanford Climate Change Facilitator
- Leads product architecture, strategy, go-to-market
- TKS Innovator | Building in public (1,747 followers)

Planned key hires: Chief Scientist (strip chemistry), VP Field Operations, Regulatory Affairs

[!TIP] The age is an asset for a moonshot competition. Two teenagers who already have a BGS-validated system, shipped 5 SaaS products, and won national hackathons. Lead with competence, let the age create the "wow" factor.

SLIDE 12 — The Ask

Purpose: Clear, decisive, no ambiguity.

Raising: \$250,000 (Pre-Seed)

Use of Funds	%
Awareness & Stakeholder Outreach	30%
Salaries & Operations	30%
Legal & Administrative Setup	25%
Technical Infrastructure	15%

What this enables:

- Put validated solution in front of governments and NGOs
- Build public awareness that safe-looking water can kill
- Begin transition from validated paper → deployed system

Future funding roadmap:

- Seed (\$1M) → Strip R&D, Bangladesh pilot, team
- Series A (\$5M) → India pilot, regulatory, manufacturing
- Series B (\$15M) → Scale to 5 countries

SLIDE 13 — 10-Year Vision (THE FUTURE)

Purpose: This is the moonshot slide. Make it audacious. Paint the world you see.

Near-term (1–3 years): Arsenic screening in Bangladesh, India, Vietnam. Prove the model. Save lives.

Mid-term (3–5 years): Planetary Hydrogeological Intelligence (PHI) platform. Any country uploads well coordinates → gets a risk map. Detect 50+ substances. 100M wells screened across 20 countries.

Long-term (5–10 years):

- **Launch our own satellite** optimized for hydrogeological remote sensing
- Become the definitive planetary water intelligence layer
- Expand: arsenic → fluoride → uranium → lithium → illegal mining → geothermal
- Parametric water quality insurance as a global financial product
- Not just testing water — **reading everything the earth is trying to tell us**

Spoken framing: *"We started by predicting which wells are toxic. We're building the intelligence layer that reads everything the earth has always been broadcasting — but nobody was listening."*

SLIDE 14 — Close

Purpose: Land the mission. Leave them with the conviction.

On screen: *"Clean water is not a privilege. It's an engineering problem."*

- Contact info
 - LinkedIn links
 - QR code to landing page or one-pager
-

Part 3: Presentation Delivery Guide

Opening (First 60 Seconds)

Don't introduce yourself first. **Open with the story.** Walk to center stage and say:

"In rural Bangladesh, a mother draws water from a tube well installed twenty years ago..."

Tell the story from Slide 2. Then pause. Then say: "My name is [Name]. My co-founder and I built a system that makes sure no mother has to wonder if her water is safe. Ever."

Pacing

- Slides 1-4 (The Problem): **Slow, deliberate.** Build tension.
- Slides 5-7 (The Solution): **Confident, energetic.** This is the turn — the audience should feel relief.
- Slides 8-10 (Business): **Steady, credible.** Data-driven. No need for theatrics.
- Slides 11-14 (Team/Ask/Vision): **Building intensity.** End on a high. The vision slide should feel like a crescendo.

Body Language (TKS: Boss Mentality)

- Stand center stage, not behind a podium
- Make eye contact with individual judges, not the slides
- Use open hand gestures, never crossed arms
- When showing big numbers, pause and let them sink in
- On the vision slide, step forward slightly — physically move toward the audience

Questions to Prepare For

VCs/judges will likely ask:

1. "How do you know the satellite model generalizes to new countries?"
2. "What's your regulatory pathway?"
3. "Who's your first customer? Do you have LOIs?"
4. "How defensible is this? Can Google replicate it?"
5. "What happens when you get a false negative?"
6. "Why should I believe two teenagers can execute this?"

For #6 specifically: Don't get defensive. Say: "We've already built and validated the system on 56,952 wells across two continents. The results speak for themselves."

Then move on.

Part 4: Design Principles

DO:

- Dark backgrounds (deep navy, charcoal) with white text — feels institutional
- One accent color: teal or cyan (water association)
- Clean serif font for headers (like Instrument Serif, Playfair Display)
- Sans-serif for body text (Inter, Outfit)
- Satellite imagery and topographic maps as visual elements
- Data visualizations that look like academic publications
- Plenty of white space — don't crowd slides

DON'T:

- Bright startup colors (neon green, orange, hot pink)
- Stock photos of "concerned scientists" or "clean water"
- Canva templates
- Gradient backgrounds with text overlaid
- More than 6 bullet points on any slide
- Clip art or generic icons
- Walls of text — you're presenting live, the slides support YOU

Text Density Rule (Hybrid Deck)

Since judges will also read this without you:

- **Headline:** Large, bold, stands alone
 - **3-5 supporting points:** Enough that a reader gets the point
 - **Speaker notes:** Put the full narrative in speaker notes for your own reference
-

Part 5: Competitive Moats to Emphasize

From your comparison doc, these are your five defensible advantages (use on Slide 6 or as a backup slide):

Moat	Why It's Hard to Copy
Data network effects	Every strip result improves Phase 1. 100M wells = unbeatable training set. 5+ years to replicate.
Health worker relationships	10,000 trained community workers. Relationships can't be copied.
Open-source + commercial hybrid	Core model open (drives adoption), premium features paid (drives revenue).
Regulatory approvals	WHO/EPA equivalence in 5 countries = 18-month head start.
Scientific credibility	Peer-reviewed, validated across two geological settings.

Part 6: Key Numbers Cheat Sheet

Keep these in your back pocket for Q&A:

Stat	Value
Wells in training data	56,952
Phase 1 accuracy	86% AUC
Phase 2 combined accuracy	98.07% AUC (US) / 97.0% (Bangladesh)
Government accuracy threshold	91%
Cost per well (Phase 1)	\$0
Cost per well (Phase 2)	\$1.50
Cost reduction vs traditional	180× / 80%
Inference speed	10 milliseconds
Bangladesh: wells	10 million
Bangladesh: traditional testing cost	\$1 billion
Bangladesh: Thalassos cost	\$3 million

Stat	Value
People protected per \$	0.075/person(vs.21.28 traditional)
Lives saved (Bangladesh, 20yr)	~200,000
Healthcare costs averted (Bangladesh)	\$997 million
TAM (addressable)	\$2.2–4.4B/year
Year 5 revenue projection	\$80M
Substances detectable	50+
Total parameters	73 (41 remote + 5 strip + 27 derived)